

Data Preparation Milestone 2

Initial Look at the Data (Value Range Check):

Before cleaning

Name	realistic_synthetic_heart_data_large.csv
realistic_synthetic_heart_data_large	25 fields 250000 rows

Field Name	Physical...	Rem...
HDL Cholesterol	realistic...	HDL...
Triglycerides	realistic...	Trigly...
Smoking	realistic...	Smoki...
Diabetes	realistic...	Diabe...
Family History	realistic...	Family...
Physical Activity	realistic...	Physic...
Alcohol Consumption	realistic...	Alcoh...
Diet Quality	realistic...	DietQ...
Previous Heart Failure	realistic...	Previo...
Kidney Disease	realistic...	Kidne...
Anemia	realistic...	Anemia...
Stress Level	realistic...	Stress...
Sleep Duration	realistic...	Sleep...
Heart Rate	realistic...	Heart...
ECG Abnormality	realistic...	ECG...
Medication Type	realistic...	Medic...
Medication Choice	realistic...	Medic...
Heart Failure	realistic...	Heart...
Heart Disease	realistic...	Heart...

After importing the **realistic_synthetic_heart_data_large.csv** file into Tableau, my first step was to review the basic values for some of the key columns. This is essentially a quick check to ensure the data falls within expected ranges.

- **Age:** Represents the patient's age. The data appears to span from approximately 18 to 95 years. This range seems reasonable for a general adult population.
- **BMI:** Body Mass Index, a measure of body weight relative to height. The values in this dataset range from about 15.0 to 55.0. These values cover the spectrum from underweight to obese, which is plausible in a diverse group.
- **SystolicBP and DiastolicBP:** Blood pressure readings. The recorded values are roughly between 90 and 220 for systolic pressure and 50 to 140 for diastolic pressure. These ranges encompass what are generally considered healthy to significantly elevated levels.
- **Blood Lipids:**
 - **Cholesterol:** Generally in the range of 100 to 400.
 - **HDL_Cholesterol:** Often referred to as "good" cholesterol, ranges from about 20 to 100.
 - **Triglycerides:** Another type of fat in the blood, between approximately 50 and 500.These ranges are clinically relevant for assessing cardiovascular risk.
- **PhysicalActivity:** Measured in minutes per week, varies from 0 to 300, representing a wide range of activity levels.
- **StressLevel:** Recorded on a scale of 1 to 10.
- **SleepDuration:** In hours, with values around 4.0 to 10.0.
- **HeartRate:** In beats per minute, generally between 40 and 120.

The categorical columns show the expected distinct values:

- **Sex:** Male, Female.
- **AlcoholConsumption:** None, Light, Moderate, Heavy.
- **DietQuality:** Poor, Fair, Good, Excellent.
- **Smoking:** 0 (No), 1 (Yes).
- **Diabetes:** 0 (No), 1 (Type 1), 2 (Type 2).
- **FamilyHistory:** 0 (No), 1 (Yes) for heart problems.
- **PreviousHeartProblems:** 0 (No), 1 (Yes).
- **KidneyDisease:** 0 (No), 1 (Yes).
- **Anemia:** 0 (No), 1 (Yes).
- **ECG_Abnormality:** 0 (No), 1 (Yes).
- **Medication_Hypertension:** 0 (No), 1 (Yes).
- **Medication_Cholesterol:** 0 (No), 1 (Yes).
- **Heart Failure:** 0 (No), 1 (Yes).
- **Heart Disease:** 0 (No), 1 (Yes).

My initial review did not reveal any obviously erroneous values or significant missing data. Tableau also appeared to correctly interpret the data types for each column.

Initial Data Quality Assessment and Minor Adjustments:

The primary goal of this initial check was to identify any immediate data quality issues. While the synthetic nature of this dataset contributed to its overall cleanliness, a closer look prompted a few minor considerations for optimal analysis in Tableau.

For the numerical variables, the observed ranges generally align with what would be expected in a diverse adult population:

- **Age (18-95 years):** This range is acceptable as it represents the typical adult lifespan and is inclusive of various age groups that might be relevant in a study focusing on heart health.
- **BMI (15.0-55.0):** This range, while broad, encompasses the clinically recognized categories from underweight to morbidly obese. It's plausible in a large, diverse dataset and allows for the exploration of BMI's impact across the health spectrum.
- **Blood Pressure (Systolic 90-220, Diastolic 50-140):** These ranges include healthy blood pressure levels as well as various stages of hypertension and hypotension. Such variability is expected in a dataset exploring cardiovascular factors.
- **Blood Lipids (Cholesterol 100-400, HDL 20-100, Triglycerides 50-500):** These ranges are within the clinically relevant thresholds for assessing cardiovascular risk. The variability allows for the analysis of different lipid profiles and their potential correlations with other health indicators.
- **PhysicalActivity (0-300 minutes/week):** This wide range reflects different lifestyle habits, from sedentary to highly active, which is important for studying the impact of exercise.
- **StressLevel (1-10) and SleepDuration (4.0-10.0 hours):** These subjective and behavioral factors exhibit a plausible range of human experiences.
- **HeartRate (40-120 bpm):** This range includes typical resting heart rates as well as elevated rates due to activity or underlying conditions.

One specific adjustment I made in Tableau, although minor, was to explicitly define the **data type** of the categorical variables that were encoded numerically. For instance, while Tableau initially interpreted 'Smoking', 'Diabetes', 'FamilyHistory', etc., as numerical measures (due to the 0s and 1s), I explicitly changed their role to **Dimensions** with a nominal (categorical) data type. This ensures that Tableau treats these variables as distinct categories for analysis (e.g., counting the number of smokers vs. non-smokers) rather than performing numerical calculations on them (which would be meaningless in this context).

Similarly, for the 'Diabetes' column (with values 0, 1, and 2), I confirmed it was treated as a nominal dimension to distinguish between individuals without diabetes, those with Type 1, and those with Type 2.

While these weren't corrections of erroneous data, they are crucial steps in **data preparation** to ensure that Tableau interprets the variables correctly for meaningful analysis and visualization. This proactive adjustment helps prevent misinterpretations later in the analysis process.

Overall, the dataset appears to be in good condition, and the minor data preparation steps taken in Tableau focused on ensuring the categorical variables are correctly identified for analysis.

After checking and cleaning

realistic_synthetic_heart_...

26 fields 250000 rows

100 rows

Name

realistic_synthetic_heart_data_large.csv

Fields

Type	Field Name	Physical...	Rem...
#	Patient ID	realistic_...	Patien...
#	Age	realistic_...	Age
Abc	Sex	realistic_...	Sex
#	BMI	realistic_...	BMI
#	Systolic BP	realistic_...	Systol...
#	Diastolic BP	realistic_...	Diasto...
#	Cholesterol	realistic_...	Chole...
#	HDL Cholesterol	realistic_...	HDL_...
#	Triglycerides	realistic_...	Triglyc...
#	Smoking	realistic_...	Smoki...
#	Diabetes	realistic_...	Diabet...

etic_heart_ase	# realistic_synthAnemia	# realistic_synthetic_hStress Level	# realistic_synthetic_heaSleep Duration	# realistic_synthetic_hHeart Rate	# realistic_synthetic_heaECG Abnormality	# realistic_synthetic_heart_data_largeMedication Hypertension	# realistic_synthetic_heart_data_largeMedication Cholesterol	# realistic_synthetic_heaHeart Failure	# realistic_synthetic_heaHeart Disease
0	0	5	4.00000	77	0	1	1	0	0
0	0	6	7.60000	73	1	0	0	0	1
0	0	6	6.90000	59	0	1	0	0	0
0	0	5	6.60000	64	0	0	1	0	0
0	0	8	7.70000	75	0	0	0	0	0
1	1	3	6.20000	67	0	1	0	1	1
0	0	1	6.20000	77	0	0	0	0	0
0	0	4	6.00000	73	0	2	0	1	1
1	0	6	9.60000	53	1	1	1	0	1
1	0	6	5.40000	78	1	1	0	1	1
0	0	5	6.20000	70	0	0	1	0	1
0	0	1	7.20000	80	0	1	1	1	1
0	0	5	6.40000	103	0	0	0	0	0
0	0	3	4.60000	58	0	1	1	1	1
0	0	6	5.50000	74	0	1	0	0	1